

Maria G Bryant
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Work Experience

Mechanical Engineer

Airbus- Mobile, AL

Jan 2024-June 2024

- Inspection and upgrading of aircraft systems and correction of any problems with design and testing.
- Wrote detailed reports on the changes.
- Tests were conducted in 2D and 3D for better illustration.

Mechanical Engineer

FedEx Ground packing Systems – Manchester, NH

June 2020 - May 2023

- Managing the smooth project of contract with customers and consumers.
- Improving working projects with training sessions.
- Collaborate with cross-functional teams to design, develop, and optimize mechanical packaging solutions for products, ensuring they meet functional, safety, and regulatory requirements.
- Ensure packaging adheres to industry standards and FedEx policies.
- Conduct rigorous testing to ensure packaging durability and sustainability.
- Enhance efficiency, reduce waste, and optimize costs in packaging processes.
- Collaborate with teams to integrate packaging into the supply chain.

Materials/Structural Engineer

UTC Aerospace Systems - Rockford, IL

January 2017 to November 2018

- Performed analysis of various flame sensing and avionics products and testing procedures for aircraft sensing systems, using PRO-E/CREO.
- Performed pre-processing FEA with CATIA on large aircraft structures.
- Designed Vibration test tube and Vibration/shock fixture for FC.

Project Management

RCF Technologies - Vidalia, GA

August 2012 to November 2016

- Managed projects with responsibilities for creating and attaining project objectives, requirements and costs.
- Performed optical, thermal, vibrational analysis with SolidWorks for project failure analysis and for developing new rotating, shaft projects, and interior systems.
- Analyzed the pumps and vacuum to determine causes of excess thermal stress.
- Performed detailed calculations on the systems in: Abacus, Math-cad, MATLAB, ANSYS Multi-physics, SolidWorks, Microsoft Office Suite, Microsoft Project.

- Performed 2 Dimensional and 3-Dimensional Design and Computations of the flow around Airfoils-using Hyper Mesh for FEA analysis.
- Introduced, modified, improved, and advanced technologies to current, new, and potential customers through presentations, training sessions, and/or demonstrations.
- Performed FEA tests on the systems and wrote detailed reports on the results of tests.
- Manage \$10M budget and 50 people.

Project Manager

BAE Systems - Hudson, NH

August 2011 to July 2012

- Performed calculations on various aircraft engines, to determine their performance, improve efficiency, depending on customer's requests. These included turbine and reciprocating aircraft engines.
- Performed detailed analysis of the interior systems and structures by applying Abaqus program.
- Verified the accuracy by performing 2D and 3D analysis of the structures with the NASTRAN & Fortran FEA program.
- Performed structural, thermal, and vibrational analysis on newly designed systems, by using SolidWorks.
- Designed and analyzed rotating and shaft systems of aircraft under Government Contract. Performed Stress and Life Analysis on Turbine Airfoils and Turbine Rotors.
- Wrote detailed reports on the results of the tests.
- Managed \$100K budget and 20 employees.

Project Engineer

Special Hermetic Products Inc - Wilton, NH

August 2008 to June 2011

Performed calculations in: Mathcad, MATLAB, Opera, Pro/E, Microsoft Office Suite, Microsoft Project.

- Analyzed and provided solutions for Design Failure Mode Effects Analysis (DFMEA) and Design for Manufacturing (DFM).
- Modified and upgraded ion chambers, advanced capacitors using Pro/E.
- Performed structural, thermal, and vibrational analysis on newly designed rotating and shaft systems.
- Performed 2D and 3D analysis with Hyper Mesh FEA Program and wrote detailed reports on the results of the tests.
- Established connections with new suppliers and vendors to ensure that final products are of the highest quality and delivered to the customer on time.

Structural Analyst

Dynamics Research Corporation - Andover, MA

August 2005 to June 2008

Performed structural, thermal, and vibrational analysis on newly designed rotating engine systems.

- Designed Versa Module Euro- Card GPS Embedded Modules to be installed on aircraft.
- Upgraded Temperature Probe for Pengelus Integrated Gyroscope project.
- Performed calculations in: Math-cad, MATLAB, Multi-physics, Microsoft Office Suite, Microsoft Project.
- Supervised data analysis to ensure final products maintained the highest quality and could be delivered on time.
- Applied FEA as part of the certification process to include fluid sensing and stress analysis.

Structural Analyst

GE Transportation - Lynn, MA May 2003 to May 2005

- Verified calculations with Pro-E on drawings and calculations.
- Project Engineer in upgrading main fuel pump and after-burner system of J85 Turbo Engine.

- Performed calculations in: Math-cad, MATLAB, ANSYS Multiphysics,
- Fluent in German. Opera, Pro/E, Microsoft Office Suite, Microsoft Project.
- Improved efficiency of main fuel pumps and maintenance instructions for turbines and anti-ice valves.
- Applied FEA with NASTRAN and/or Fortran as part of the certification process to include fluid sensing and stress analysis.

Mechanical Engineer

Kimball Physics Inc - Wilton, NH

January 2001 to February 2002

- Designed Time Tracking Systems, Electron and Ion Guns, Emitters, and Detectors using Pro-E. • Upgraded UHV-Chambers, Modular Components, produced for NASA.
- Developed Gun Housings for the Electron and Ion Guns.
- Supervised Custom Vacuum Chamber drawings to ANSI Y14.5M Standards.

Mechanical Engineer

Meggitt Avionics Inc - Manchester, NH

September 1997 to January 2001

- Tested air curtain and pressure transmitters for turbo-machinery using PRO-E. • Evaluated Pressure Transmitter to determine causes of surface corrosion.
- Designed Vibration test tube and Vibration/shock fixture for Flame Characterization Detector installed into F-22 engine, Flame-Out Detection Systems installed in gas turbine engines with PRO-E.
- Upgraded Power supplies for Flame-Out Detectors, Air curtain for Smoke Detectors.
- Low Level Fuel Sensor for jet engine to customer specifications, Oil quantity transmitters for transport aircraft, Interfaces for flame characterization scanners.

Education

BS in Mechanical Engineering

University of New Hampshire - Durham, NH 1997

AS in Aeronautical Engineering

Daniel Webster College - Nashua, NH 1994

General Drafting

North American School of Drafting - Scranton, PA

1993

BS in Aviation Management

Daniel Webster College - Nashua, NH 1992

Skills

KEY SKILLS

- Tests and inspections of products, services, or processes to evaluate quality and performance.

- Effective verbal and written communication
- Budget development, Cost Analysis
- Project Planning & Completion
- Design Documentation
- Fluent in German

TECHNICAL EXPERTISE

ANSYS 15 years,
 Multiphysics 12 years,
 AutoCAD 18 years,
 Autodesk Inventor 14 years,
 Hyper 10 years,
 IPC STDs 10 years
 Mechanical Desktop 15 years
 Mil STD 810 12 years ,
 Mil STD 461 12 years ,
 Pro-E/CREO 15 years ,
 SolidWorks 15 years,
 Siemens NX 12 years,
 Mathcad 16 years,
 MATLAB 14 years,
 Microsoft Office Suite 14 years,
 Microsoft Project 14 years,
 Algor 15 years ,
 Excel 15 years,
 WordPerfect 15 years,
 GD&T 18 years ,
 ASME Y14.5 M and Y 14.5.1 M 18 years,
 Abaqus 12 years,
 FEA with NASTRAN & FEMAP 15 years.
 Word 16 years